

Dehydration in Elderly: Revisiting the Assessment and Management Strategies

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ABSTRACT

Dehydration, like many other aspects of an aging body, is often neglected by patients and physicians alike. Not only does it sometimes become difficult to clinically assess and identify dehydration in the elderly, but it also becomes difficult to attribute gross changes in functioning to something as simple as water depletion. This can be counterproductive to the overall health and even survival of elderly patients if diagnosis is delayed. We propose a comprehensive hydration stewardship program, with public health and clinical interventions, to prevent dehydration and its complications in vulnerable segments of society, such as the elderly. This short review summarizes current evidence available for the diagnosis and identification of dehydration in the elderly and shares preventive strategies to prevent its occurrence and complications.

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DEFINITION

Dehydration is a multifactorial and multifaceted syndrome, that is, a constellation of symptoms, signs, and laboratory abnormalities, which may be organ or life-threatening. There is no universally accepted definition for dehydration.¹ For all practical purposes, it is defined as "depletion in total body water (TBW) content caused by pathological fluid losses, diminished water intake, or a combination of both."

EPIDEMIOLOGY

The burden of dehydration among older adults has been described across the world using different nonstandardized diagnosing criteria. These include classical clinical findings, biomarkers [like blood urea nitrogen (BUN) to creatinine ratio, electrolytes, and specific gravity], and measurements of the adequacy of fluid intake.² Paulis et al., in a systematic review of nineteen studies, found wide variation in various prevalence studies, mainly due to differing definitions of dehydration used by authors.³ This suggests a lack of standard consensus criteria to diagnose such cases, more so in the elderly population. The lack of a standard operational definition for dehydration prevents the comparability of research findings and prevalence results.

Lesnik et al. studied 410 patients, 65 years and older from Slovenia, where they used biochemical parameters like osmolality, electrolytes, and BUN to creatinine ratio to screen for dehydration.⁴ The prevalence of dehydration ranged from 51 to 41.3%, depending on whether the patient was institutionalized or in the community,

respectively. Schettino et al. carried out a systematic review of the literature associated with stroke, dehydration, and dysphagia and found prevalence of dehydration in poststroke patients during hospitalization was up to 66%.⁵ They also noted that dehydration, defined using the biochemical parameters, was associated with the development of venous thromboembolism during the course of their stay in the hospital. This reinforces previous studies that show dehydration rates worsen rather than improve after hospital admission, suggesting a strong need for effective hydration stewardship.

The main causes of death among the elderly during high ambient temperatures include severe dehydration, heat-related pathologies (hyperthermia, heat syncope, and heat stroke), acute cardiovascular events, including heart failure in preexisting compromised heart patients, prerenal failure, the rapid development of delirium, and subsequent chest infections, with one organ failure precipitating multi-organ failure, characteristic of geriatric patients.^{6,7} Thus dehydration becomes a cause of mortality in these patients. This is precipitated by impaired thermoregulatory mechanisms, that is inevitable with aging. The elderly population, especially those with premonitory frailty, with or without underlying debilitating comorbidities, and use of dehydration-precipitating drugs like diuretics or sedatives, had the highest death rate in France and Italy during the 2003 heat wave. The age-associated physiology makes it even more important for the caregiver to get involved in the hydration maintenance of such individuals before any complication occurs. Once hospitalization is required

occurs, bringing the patient back to the premonitory state of functioning becomes quite a daunting task for any healthcare facility.^{6,7}

Salam et al., in a 4-year prospective study, observed the effects of meteorological factors and their association with seasonal stroke risk.⁸ They demonstrated a seasonal pattern in the incidence of stroke with a higher risk of both ischemic as well as hemorrhagic stroke during the summer months, with higher solar radiations, irrespective of temperature rise. Such studies point to heat-related mechanisms in addition to dehydration in the causation of these events.

ETIOPATHOGENESIS

Older individuals are at increased risk for dehydration, and multiple reasons converge to increase this risk. These are listed in Table 1.

Several studies prove this by demonstrating that in healthy elders, even if the water and temperature regulating mechanisms function normally in the controlled, nonvarying environment, they become impaired as soon as the excessive physical stress of temperatures ensues. At these extremes, compensation is ineffective, and this becomes apparent earlier in elders with chronic comorbidities. Elders, with increasing comorbidities, are highly vulnerable, and this lands them with various geriatric giants, which increases their risk of acute diseases and mortality.^{9,10}

SCREENING FOR DEHYDRATION

Screening can be done using clinical and/or biochemical parameters. Validated scales have been created for use in ambulatory as

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well as hospitalized patients by medical and nursing staff. Some useful screening tools are listed in Table 2.

Screening Tools^{11–15}

The orthostatic blood pressure changes and skin turgor, though good clinical makers of acute dehydration, are of less diagnostic value in chronically dehydrated elderly and thus are not included in most of these screening tools.

DIAGNOSIS OF DEHYDRATION

Lacey et al. published a multidisciplinary consensus statement on the definition and diagnostic methods of dehydration, and the key findings suggested by them were that there is no universally accepted definition for dehydration.¹⁵ For all practical purposes, dehydration is defined as “depletion in TBW content caused by pathological fluid losses, diminished water intake, or a combination of both.”

Direct measurement of plasma osmolality is the gold standard for determining dehydration. Underhydration is a precursory condition to dehydration associated with a more insidious onset of clinical features but a similar poor outcome.

Changes in skin turgor, sunken eyes, and dry mucosal membranes are of more clinical value in younger individuals. For older adults, due to age-related changes in skin and mucosa, the diagnostic value of such clinical findings becomes less marked. Thus, making

a diagnosis, and more so, guiding fluids based on skin/mucosal changes, would not be recommended.

Assessing skin turgor at the sternal level is widely used, but like measurement of skin turgor at any other level might become comprised due to age-related changes in skin tissue. From a systematic review, Hooper et al. found that the clinical symptoms of “expressing fatigue” and “missing drinks between meals” were the only measures with

the ability to identify those at greatest risk of impending dehydration.¹²

Table 3 shows various methods of diagnosing hydration status in the elderly with their merits and demerits.

Studies have found urine color and specific gravity to be easy, low-cost, and effective indicators of hydration status in older individuals with adequate renal function. They can be used as useful measures to monitor hydration in nursing

Table 3: Diagnosis of hydration status

Different methods to diagnose dehydration in elderly	Merits	Demerits
Serum osmolality	Gold standard recommended by European Society for Clinical Nutrition and Metabolism.	
Serum sodium		Only slight alteration in mild dehydration.
Urine specific gravity Urine osmolality Urine output Urine color charting	Easy and reliable. Self-monitoring at home is possible.	Affected by comorbidities and drugs.
Saliva flow rate Saliva osmolality		Difficult to collect and send Diurnal variations.
Clinical signs: Skin turgor/ Sunken eyes/ Mucosal membranes	Early sensitive indicators as compared to blood parameters. Skin turgor is best assessed at the sternal level in the elderly.	Age-related changes in skin elasticity and poor oral health make these less reliable in elderly.
Combination of clinical and lab parameters	Best recommended approach.	

Table 1: Etiopathogenetic factors of dehydration

- Lack of thirst due to changes in hypothalamic thirst receptors as well as peripheral baroreceptors, which diminishes their ability to sense volume deficit.
- Differences in body composition, including loss of muscle tissue and increase in fat content.
- Poor thermoregulatory response to rise in environmental temperature changes.
- Changes in kidney physiology, including age-related loss of regulating fluids and ability to handle excess sodium.
- Elderly individuals have cellular water content, which jeopardizes thermoregulation through less sweat generation and a diminished ability to dissipate body heat via dehydrated skin.
- It also depends on the underlying chronic hydration status of the patient, as those with some form of chronic water diminution will more likely to be affected by acute water loss, which in turn depends on drinking habits, cognition, and the involvement of caregiver in overall maintained of patients general health.^{8,9}

Table 2: Screening tools for dehydration

- The dehydration risk appraisal checklist keeps the risk population at higher score, including those >85 years of age, females, those with significant medical/or neuropsychiatric conditions, poor hydration behaviors, those taking medications with risk of diuresis or water loss, and also includes biochemical parameters like urine osmolality, urine color, hematocrit, BUN/creatinine ratio, serum sodium, serum osmolality.
- Dehydration screening tool (DST) designed by Vivanti et al. Gives a fair idea about hydration in hospitalized elderly people. It is composed by the most promising 11 items based only on history taking alone.⁴
- The geriatric DST, a comprehensive toolkit, is divided into three sections:
 - Section 1: Demographics linked to dehydration (age/gender/education/comorbidities).
 - Section 2: Physical signs of dehydration (tongue hydration/weight-body mass index/axillary hydration).
 - Section 3: Lab parameters (hematocrit/blood urea/serum osmolality).
- The Northumbria assessment of hydration is a simple, nurse-guided screening tool to identify and stratify elderly patients according to their risk of dehydration, which can be easily used for daily charting by nursing staff in the ward.

homes (by nurses) and at home (by patient and his/her caregivers).¹¹

Limitations in using urine indices to estimate specific gravity include the following:

- Medications and foods have the ability to change urine color.
- Motivation of staff or caregiver and ability of the patient to diligently monitor or chart correctly identified findings of urine color.
- Compromised renal function in chronic kidney disease.
- Timing of collection of urine.

In a recent study of younger adults, Perrier et al. found that a late afternoon (between 4:00 and 8:00 p.m.) specimen best reflected overall hydration status as measured by urine osmolality when compared to a 24-hour urine collection. This would need to be tested with a sample of older adults. Rodrigues et al. found a 24-hour urine osmolality was negatively associated with hydration score.¹² Johnson et al. created a composite index, the fluid retention index, which includes urine color, specific gravity, and urine creatinine. They proposed that the renal system is more responsive to water deficiency as compared to blood levels, and thus urine indicators are better than serum for biochemical diagnosis of dehydration.

Salivary osmolality is coming up as a novel indicator of dehydration, both in water and water-solute type of dehydration, the only limitation being sample collection on a regular basis for monitoring, and those patients with poor health, in which case, the sensitivity of the test might get affected.¹

Another very good modality being used in the geriatric clinic is bioelectrical impedance analysis (BIA), which has the ability to screen for sarcopenia/muscle mass, body composition, and total water content in the body. Upcoming research may strengthen the utility of BIA in both outpatient and inpatient settings.

Hooper et al. demonstrated that serial measurement of history of fatigue grading, the number of missed drinks between meals, and BIA resistance of 50 hz has good sensitivity in diagnosing impending

and ongoing dehydrated patients. These simple tests become more important at the community level for timely diagnosis of dehydration and appropriate education so as to prevent decompensation and disease.

DEHYDRATION PREVENTION STEWARDSHIP¹⁶⁻¹⁸

Dehydration should be considered a chronic disease, and water intake a medical therapeutic priority in geriatric practice. Preventing dehydration, or managing it in its early stages, is the key to ensuring geriatric health. Prevention of dehydration can be ensured by appropriate public health messaging, as well as counseling and clinical interventions.

Defining Adequate Daily Fluid Intake Goal

An evidence-based nursing review concluded that all older adults should have an individualized fluid goal determined based on a documented standard for daily fluid intake. Numerous standards have been developed, several based on liquid intake, while others are based on water intake that is inclusive of food and fluid. [Gaspar et al. provide a rationale for the use of the following formula as a recommended water intake standard \(from food to fluid\): \$1600 \text{ mL/m}^2 \times 0.75\$. This formula provides the most individualized approach to the establishment of a water intake recommendation and should be attainable.](#) The National Institute of Nutrition, Hyderabad, has also recommended daily allowances for water intake in various age groups I, II, and III.

Dehydration Preventive Strategies: Current Evidence

At the National level

At the government level, taking cues from various European countries, a mass information dissemination system must be put in place. This can include electronic alert notices, early warning systems, and mass media use to make dehydration stewardship a priority. Mitigative strategies like urban design modifications ([Table 4](#)) to include more green spaces, reflective material use on the road, and

clothing and dietary advisory for the elderly must be implemented. Extensive preventive planning is needed to translate scientific evidence base into action at a mass level.

At the Community Level

Community plays a huge role in the overall well-being of the older adult, whether it involves the dissemination of vital information regarding good hydration practices or during the time of emergencies, where support from the hospital might be essential. Individual outreach programs, where a home visit to every elderly person for regular monitoring of well-being, should include a short history of water intake practices. Such initiatives might seem very nonspecific, and the necessity of such measures might seem quite futile to physicians, but such preventive measures form the basis of future medicine. Prevention might actually be a better cure: the elderly are made to realize the importance of drinking water, and dehydration and its downstream complications can be prevented.

At the Healthcare Level

Physicians, both generalists, and specialists, should be trained in recognizing and promptly treating such cases. At the index outpatient visit, they should advise the elderly of suitable water intake amounts based on comorbidities, complications, and concomitant medications and guide the patient so that he/she becomes a champion of his/her own fluid needs. For those with severe frailty, caregivers should be properly instructed to monitor urine color and common signs of poor hydration status such that management can be appropriately instituted at the home level itself, obviating the need for hospitalization. Simple techniques like regular monitoring of weight and urine color diary can also go a long way in improving the overall quality of life.

Drugs that can affect the thermoregulatory compensation, like β -blockers, diuretics, and sodium-glucose cotransporter-2 inhibitors, should be titrated accordingly. Like in any frail elderly, drugs that affect cognition should be omitted whenever feasible. Kidney disease and hydration in itself is a very specific issue, where

Table 4: Environmental modifications and hydration stewardship

- Use of air cooling systems like air-conditioners (ACs) or water coolers might be very beneficial for individuals per se. From an environmental perspective, water coolers have an edge over ACs.
- Improvement in civil engineering design of building materials to include more insulation materials and urban home designing to promote cooling, with special emphasis on roof flooring.
- Paint should be reflective and free from toxins, with regular repainting to ensure continued insulation.
- Using appliances that pollute less and emit less heat.
- Everyone should be promoted to use public transport, and these should be made accessible and affordable to the elderly to promote better utilization.

a very attentive approach to intake, output, and insensible losses should be observed carefully; a similar approach is required for heart and hepatic conditions, which have an easy tendency of decompensating or worsening if a tight fluid control strategy is not followed.

AUTHORS CONTRIBUTION

Minakshi Dhar conceived the idea, formulated the methodology, wrote the protocol, and edited the final manuscript; Kartik Mittal did the review search and helped in writing the review; Monika and Sanjay helped in the review of studies, edited the manuscript, and all authors approved the manuscript.

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